CSCY413 - Special Topics in Cybersecurity
Project Proposal

Group Members (**SID**): Haneen Gouda (**23000737**) - Mai Mudathir (**23000805**)

Project Title: PoisonGuard: ML-based Data Poisoning Defense for IDS Project.

Project Overview: The project was chosen because it examines risks in healthcare ML-based networks and how to protect IoMT (**Internet of Medical Things**) networks using ML-based IDS (**Intrusion Detection Systems**) against data poisoning attacks such as Label Flipping and Backdoor attacks, while proposing and testing a defence method that maintains detection reliability.

Problem Statement: IoMT networks depend heavily on their training data, which is collected from multiple devices and sources; as it's an ML-based system, this provides an opening for attackers to manipulate the data before training, making the model learn the wrong patterns, resulting in poor accuracy and performance, and failing to detect real attacks. In IoMT, such failures can expose patients' data, disrupt care devices, readings, or monitoring, and violate the CIA triad (**Confidentiality - Integrity - Availability**).

Aim: To evaluate the impact of a data poisoning attack on ML-based IDS in IoMT networks and elaborate a defence method to restore its detection and reliability performance.

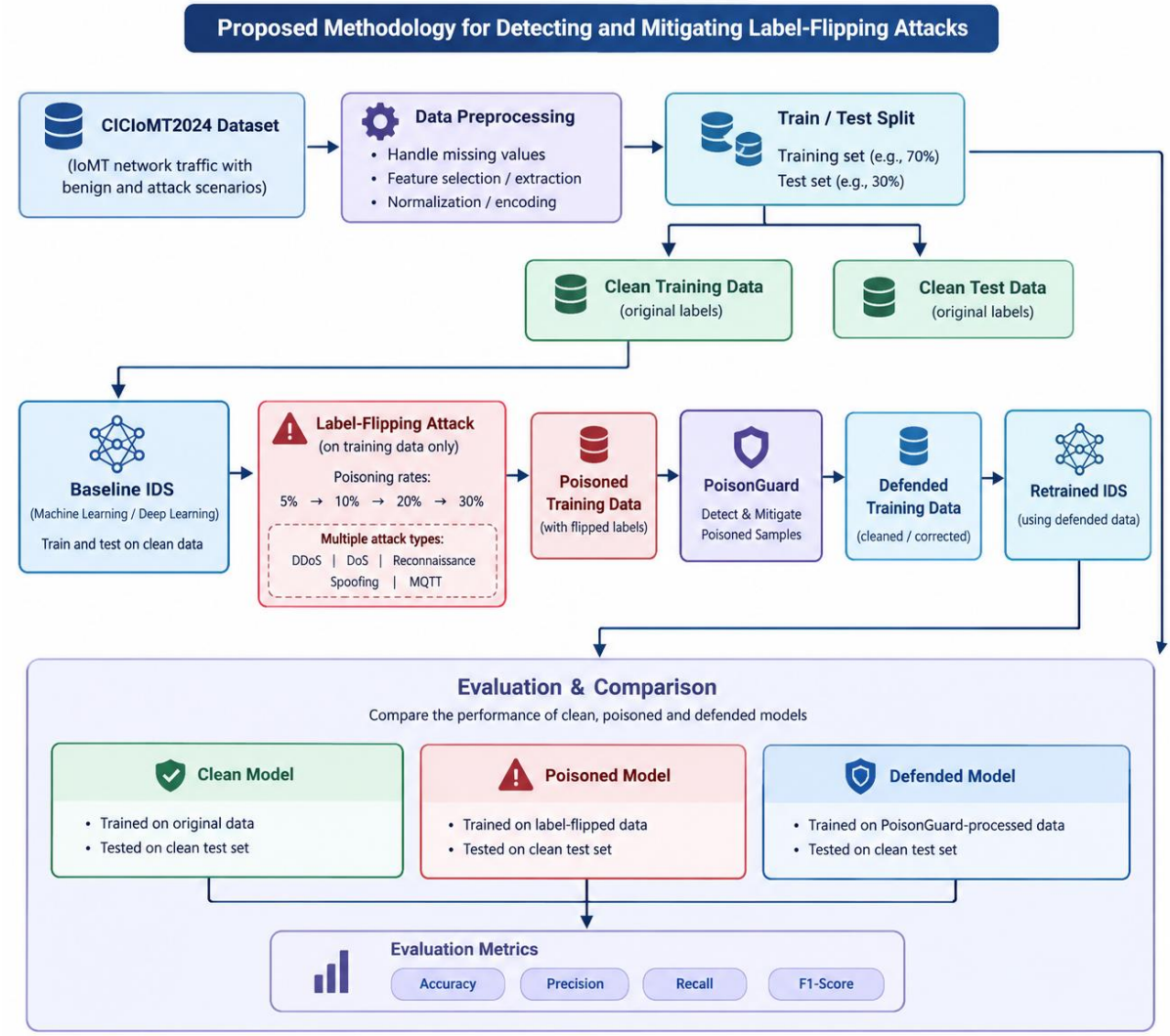
Objectives:

1. Examine current studies on IoMT security, data tampering, and ML-driven IDS protections to pinpoint the research deficiency.
2. Simulate label flippings attack at different poisoning rates.
3. Design and implement a defense method.
4. Compare the models' (baseline - poisoned - defended) classification evaluation metrics such as accuracy, precision, recall, and F1-score.
5. Discuss findings, limitations, and future enhancements and recommendations for real-world IoMT implementations.

Dataset: The dataset CICIoMT2024 will be used in our project. It is a recent Cybersecurity dataset on IoMT that includes 8.8 million network flow records from 40 IoMT (Internet of Medical Things) devices with benign/normal traffic and different types of cyberattacks such as DDoS, DoS, Reconnaissance, Spoofing, and MQTT-related attacks.

The initial attack traffic will be taken as the basis for IDS construction. The second step will be the simulation of the label-flipping attack through modification of the labels in the training set only, whereas the test set will remain untouched for reliable evaluation.

Methodology: The baseline machine learning/deep learning IDS will initially be implemented and tested with the initial data set. Label flipping poisoning impact will next be evaluated at various poisoning rates, for example, 5%, 10%, 20%, and 30%, and for multiple attack types. Lastly, the proposed PoisonGuard defense will be utilized to detect and mitigate the poisoned data, followed by the comparison of the performance of clean, poisoned, and defended models, which will be done based on accuracy, precision, recall, and F1-score metrics to determine the effects of the label-flipping attack and the efficiency of PoisonGuard.



Research Questions:

1. In what way does label-flipping poisoning influence the performance of the intrusion detection model loMT?
2. How does the growth of the poisoning ratio influence IDS performance?
3. Is there any difference in the impact of label flipping for different loMT attack types?
4. Is it possible to detect potentially poisoned training samples?
5. To what extent is it possible to protect against label-flipping attacks using PoisonGuard?

Expected Contribution & Results (Initially): The project aims to experimentally study the effect of label-flipping vulnerability on loMT Intrusion Detection Systems and explore ways to combat such attacks in the training dataset. An important part of this research will involve studying the effects of label-flipping across various categories of attack and at various rates of poisoning. This project is expected to prove that label flipping attack may adversely influence the functioning of the IoMT intrusion detection system. The degree of impact will depend on the rate of poisoning and the type of attack. PoisonGuard aims to detect and counteract the effect of the poisoned training dataset and enhance IDS performance over the poisoned model.

References:

- **Dataset link source**: Kaggle, CIC IoMT 2024 WiFi MQTT